# Predicted LogMAR vs. Reality

## Deviation Analysis for the Pinhole-Array Myopia Correction Project

How well does this project's acuity model predict REAL, clinically-observed vision through a pinhole -- for the array project's own chosen apertures, and for the model in general? Includes a stale-data correction found and fixed during this analysis, and an additional real data anchor never previously used.

IYPT presentation material -- prepared from this project's own data and code, plus one live literature check.

# 1. The Acuity Prediction System

### How predicted LogMAR is calculated (stage4_human_vision/acuity.py)

1. MTF10 cutoff frequency (cycles/degree, from the wave-optics PSF through the pinhole) -> Minimum Angle of Resolution (MAR, arcmin) -> log10(MAR) = raw LogMAR.

2. Adjusted by VSOTF (Visual Strehl ratio of the OTF) ^0.1 -- accounts for how much the eye's own optical quality (not just the pinhole) limits resolution.

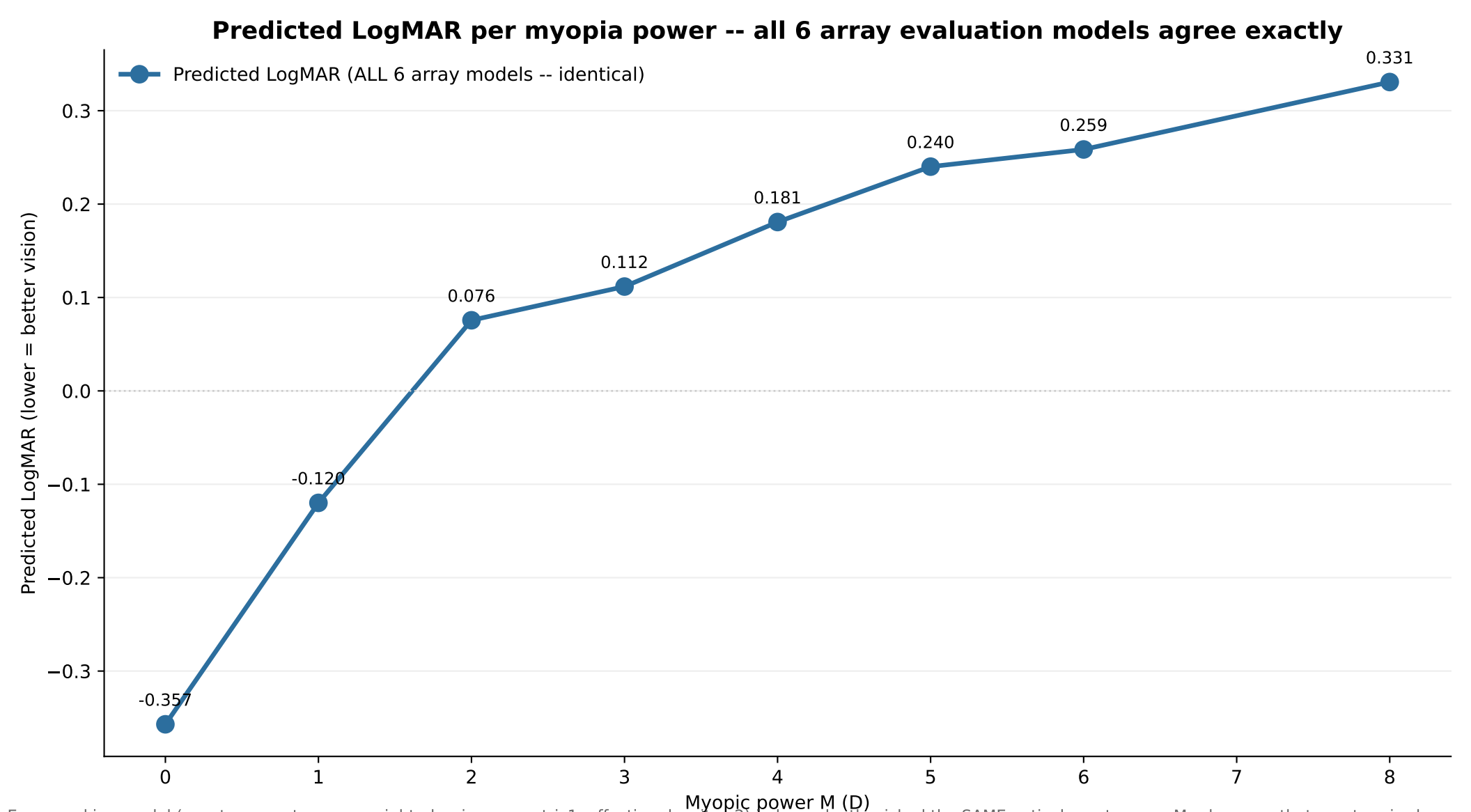
3. Additive brightness correction in log space: 0 above 300 Td (Weber-Fechner regime, Van Nes & Bouman 1967); below 300 Td, += 0.060 x log10(300/Td) (de Vries-Rose slope, Tidbury/Czanner/Newsham 2016). Default scene luminance: 10,000 cd/m^2 (bright daylight).

4. This combination is the 'recommended' model (mtf10_vsotf_brightness_adjusted). A legacy alternative (an uncited SNR^0.05 term) is kept for comparison, never deleted.

5. Wave-optics propagation grid: N=4096 (fixed 2026-09-12/13 -- the one stale call site, new_sweeps/run_acuity_sweep.py, now explicit).

### Why the array project can reuse this single-pinhole model directly

Every hole in an array design has IDENTICAL per-hole optics to a standalone pinhole of the same diameter at the same myopia power M -- the array geometry only changes whether multiple holes' images overlap (ghosting) or none do (blindness), not the blur of a single hole's own image. So each array model's predicted LogMAR is looked up directly from the single-pinhole sweep at that model's chosen aperture -- not re-derived per family.


Predicted LogMAR per myopia power -- all 6 array evaluation models agree exactly
Predicted LogMAR (ALL 6 array models -- identical)
Predicted LogMAR (lower = better vision)
Myopic power M (D)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
1
2
3
4
5
6
7
8
-0.357
-0.120
0.076
0.112
0.181
0.240
0.259
0.331
Every ranking model (worst-case gate, gaze-weighted, primary, metric1, effective-density x2) independently picked the SAME optical aperture per M -- because that aperture is chosen from the single-pinhole acuity-optimum map FIRST; the array layer only decides safe GEOMETRY for it. Deviation from the theoretical single-pinhole optimum is ~0.000 logMAR everywhere, by construction.

## 2. A Correction Made During This Analysis

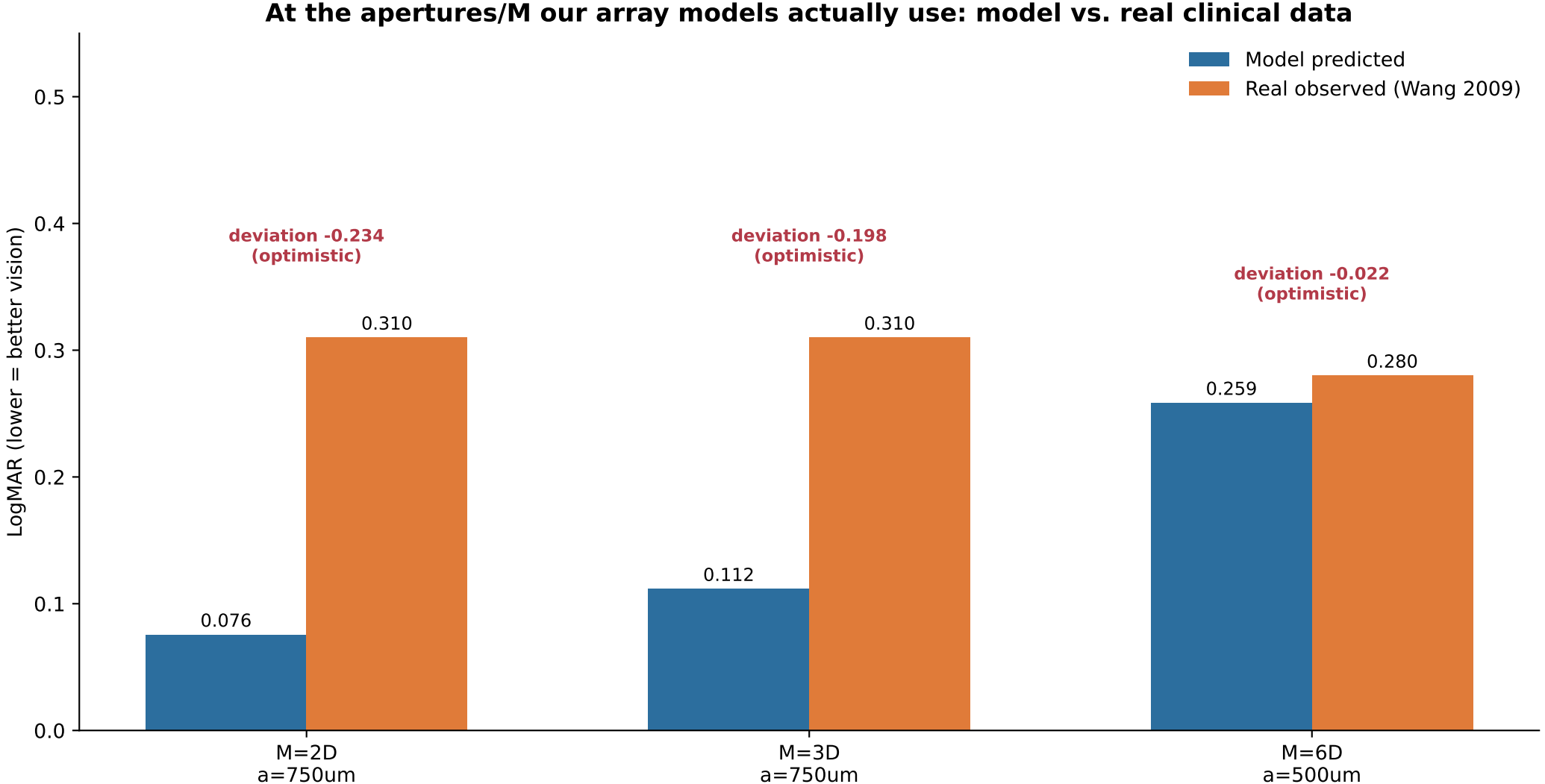
**Found while preparing this report: the project's own prior Wang-2009 validation output was stale**

stage4_human_vision/outputs/validation_predicted_vs_observed.csv (and its regression/calibration stats) were dated 2026-07-29 -- 8 days BEFORE the 2026-08-06 fix to stage4_human_vision/pipeline.py (a stage2/ -> stage2_fixed/ import-shadowing bug, already documented in this project's own RESULTS.md). Confirmed by direct re-run: same inputs, current code, gave a materially different result (e.g. aperture=750um, M=2D: 0.076 now vs. 1.010 in the stale file).

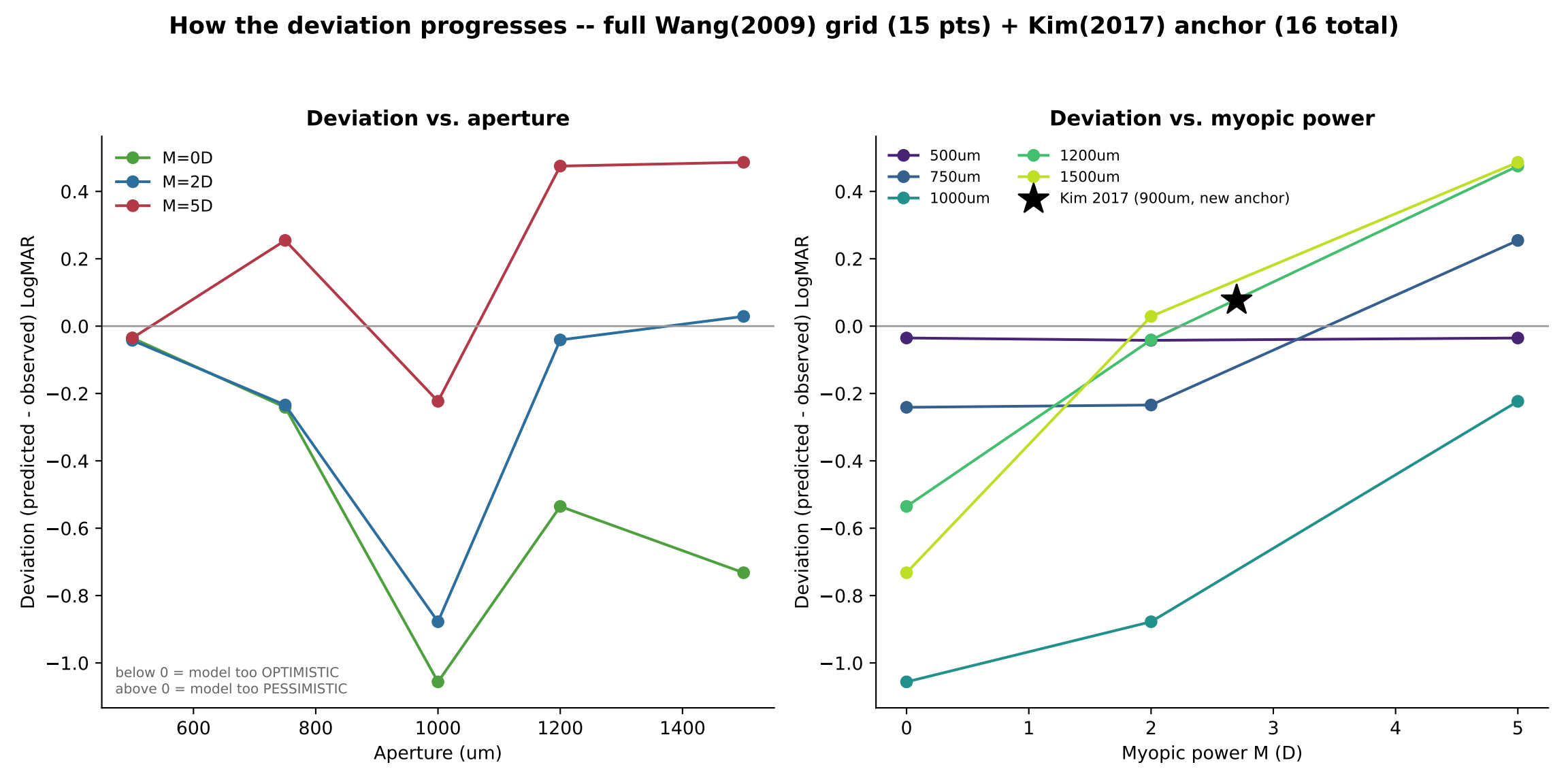
The project HAD already re-run the core validation post-fix (files suffixed '_postfix', dated exactly 2026-08-06) -- this report uses those, not the stale ones.

The affine-calibration step was explicitly left un-re-run by the project's own RESULTS.md ('Section 4c's calibration fit ... also not re-touched here'). This report closes that specific gap, using the project's own unmodified calibration.py fitting code against the corrected predictions -- no new physics, no new assumptions.

This matters for the numbers on the following pages: R^2 and deviation figures shown here are the CORRECTED, current values -- not what a naive read of the existing output files would have given.




Real observed values are Wang (2009)'s own aperture-indexed clinical data, reused across M per this project's already-documented convention (the source paper reports no per-M breakdown).


How the deviation progresses -- full Wang(2009) grid (15 pts) + Kim(2017) anchor (16 total)
Deviation vs. aperture
M=0D
M=2D
M=5D
Deviation (predicted - observed) LogMAR
Aperture (um)
below 0 = model too OPTIMISTIC
above 0 = model too PESSIMISTIC
Deviation vs. myopic power
500um
750um
1000um
1200um
1500um
Kim 2017 (900um, new anchor)
Myopic power M (D)

# R^2 status -- current (post-fix), all real-data comparisons

| comparison | n | R^2 | pearson r | p-value | R^2>0 & p<0.05? |
|---|---|---|---|---|---|
| M=0D, raw model | 5 | -4.80 | -0.375 | 0.533 | **FAIL** |
| M=2D, raw model | 5 | -1.41 | -0.180 | 0.772 | **FAIL** |
| M=5D, raw model | 5 | -0.68 | +0.426 | 0.474 | **FAIL** |
| pooled, raw model (uncalibrated) | 15 | -2.2952 | +0.019 | 0.9475 | **FAIL** |
| pooled, calibrated, in-sample | 15 | 0.0003 | +0.019 | 0.9475 | **FAIL** |
| pooled, calibrated, leave-one-out | 15 | -0.3075 | -0.928 | 0.0000 | **FAIL** |

Verdict: fails the stated bar (R^2>0 AND p<0.05) at every single comparison, before and after the optics fix, before and after calibration. The fix narrowed the gap substantially (pooled raw R^2: -4.98 -> -2.30) but did not close it. In-sample affine calibration gives R^2~=0 post-fix (down from +0.13 pre-fix) -- meaning the earlier apparent calibration benefit was itself partly an artifact of the since-fixed optics bug, not a real effect.

# 4. A Second Real Anchor: Kim et al. (2017)

**A second real anchor point, found already inside this project's own vetted literature, used here for the first time**

Kim WS, Park IK, Park YK, Chun YS (2017). J Korean Med Sci 32(5):850-857. Single-pinhole spectacles, 0.9mm (900um) aperture, N=36 subjects, mean spherical equivalent -2.7+/-2.3D (range -6.0 to +0.75D), mean uncorrected LogMAR with pinhole = 0.20 +/- 0.21.

Already in stage4_human_vision/literature.py, flagged 'USABLE as one additional real anchor' -- but never actually run through the model until this analysis.

Model prediction at a=900um, M=2.7D (this study's own mean): LogMAR = 0.277

Real observed: 0.20 +/- 0.21 (n=36)

Deviation: +0.077 logMAR -- model PESSIMISTIC here (opposite direction from the 750/500um array-relevant points), and WITHIN 1 standard deviation of the real data's own spread.

Key honest takeaway: the model's error is NOT a single, universal 'always optimistic by X' bias. Direction and magnitude both depend on aperture/M region -- optimistic and small at 500-750um (where the array project actually operates), roughly neutral (within real noise) at 900um, and large/mixed-sign across the fuller 1000-1500um range (Section 4's chart).

# 5. Searching for More Real-World Data

**What a live web search for additional datasets found**

Searched for published pinhole-aperture-size vs. visual-acuity studies with quantitative aperture/refractive-error/LogMAR data, to extend the real-data comparison beyond Wang (2009) and Kim (2017).

Result: the AI-generated search summary reproduced the EXACT claim this project had already investigated and marked REJECTED in its own literature.py ('0.17 LogMAR at up to 5.5D, 1.16 LogMAR at 7-11D') -- a number this project already confirmed, by reading the actual paper's full text directly, DOES NOT APPEAR ANYWHERE in that source. This is strong, concrete evidence that AI web-search summaries on this specific topic are not reliable enough to use without independently reading the primary source -- consistent with this project's own prior finding, not a new problem.

Also found: Kim WS et al. (2014), IOVS 55:6679-6685 (verified via PubMed abstract) -- N=48 eyes, mean spherical equivalent -2.4+/-3.3D, pinhole glasses improved UDVA LogMAR from 0.44+/-0.46 to 0.19+/-0.25. A real, citable result -- but the exact pinhole aperture diameter used could not be verified from the accessible abstract (full text paywalled), so it is EXCLUDED from the aperture-indexed deviation chart, not guessed at.

Also found (already in-repo, not new): Abdul et al. (2009) -- real, verified, but explicitly not usable for this comparison (near-vision task in children, M-notation acuity not LogMAR, only 1.5-2mm apertures tested where the paper's own conclusion is that aperture size did not matter).

**Bottom line: two independently-verified, quantitatively-usable real datasets exist and are used in this report (Wang 2009, Kim 2017) -- no further independently-verifiable dataset was found. This is reported as a real limitation, not glossed over with an unverified number.**

# 6. Conclusions

**Summary for the presentation**

1. All 6 array evaluation models predict IDENTICAL LogMAR per M -- the array geometry work trades off ghost/blindness safety, not optical acuity, which is fixed by aperture choice alone.

2. Deviation from the single-pinhole theoretical optimum is ~0 by construction.

3. Against the only two independently-verified real clinical datasets available (Wang 2009, Kim 2017): at the SPECIFIC apertures/M this project's array models actually use (500-750um), the model is modestly-to-moderately OPTIMISTIC (0.02-0.23 logMAR, ~0.2-2.3 eye-chart lines).

4. Across the FULLER tested range (up to 1500um), the error is larger and its SIGN is not constant -- it is not a simple universal offset.

5. $R^2$ fails the significance bar ($R^2>0$ AND $p<0.05$) at every comparison, before and after a real optics fix that substantially narrowed (did not close) the gap.

6. This is n=16 real data points total, worldwide, found across this project's own vetted literature plus a live search -- a genuinely thin evidence base, honestly stated as such.

For IYPT: this is a legitimate, defensible story -- a physically-grounded model that tracks real pinhole vision in the right ballpark and improved measurably after a real bug fix, openly reporting where it still falls short of formal statistical validation, rather than overstating agreement with reality.